Week 22: Review, Present, and Choose with Purpose

A course review, a public-speaking toolkit, and the opportunity cost of a research choice

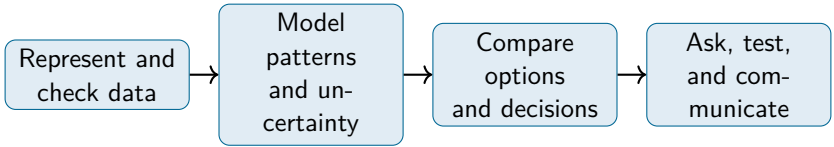
Instructor / Mentor

Kiki Research Training Program

Week 22 of the Research Training Program

Review 1: The Course Is a Research Loop

We began with ways to represent information, then learned how to analyze it, make decisions, and design a project.



The habit to keep

Start with a question and evidence. Use the simplest tool that can help answer it, then state what the answer can and cannot support.

Review 2: Data Needs a Representation

- **Week 1 — vectors and code:** variables, lists, and tables represent real measurements; a vector can describe one user's pattern.
- **Weeks 2–3 — networks and databases:** nodes and edges describe connections; tables and queries help find records that meet clear conditions.
- **Week 4 — matrices:** a matrix can represent many relationships or transform a vector, as in $Ax = b$.

Ask before calculating

What does one value, row, node, or edge mean? Which units and assumptions are attached to it?

Review 3: Change Has a Shape

Weeks 5–6: change and evidence

- A derivative describes a rate of change; an integral accumulates change.
- A research abstract can be unpacked into Problem, Method, and Result.
- A claim is only as strong as its evidence and assumptions.

Weeks 7–8: signals and structure

- The Fourier view describes a signal using frequency components.
- A graph Laplacian connects a network's structure to how values vary across it.
- A smooth-looking pattern is not automatically a useful or true one.

Review 4: Decisions Need Rules and Trade-offs

- **Week 9 — optimization:** choose values that minimize or maximize an objective while respecting constraints.
- **Weeks 10–11 — game theory:** decisions depend on what other people may choose; cooperation raises questions of fair benefit-sharing.
- **Week 12 — probability and risk:** variation and uncertainty matter, especially when rare outcomes can have large consequences.

A useful decision statement

"Given these goals, constraints, and uncertainties, this option is preferable to these alternatives for this reason."

Review 5: Regression Predicts a Number

- **Week 13:** linear regression finds a weighted relationship that can predict a continuous value, such as electricity use.
- Mean squared error summarizes squared prediction errors; R^2 describes fit relative to a simple reference.
- A score alone does not tell us whether a model is useful for a real decision.

Always compare

Test the model against a simple baseline, use data the model would really have at prediction time, and report what the errors mean in the original units.

Review 6: Classification Predicts a Category

- **Week 14:** logistic regression turns a weighted score into a probability; a threshold turns that probability into a class.
- A confusion matrix counts correct and incorrect positive and negative predictions.
- Accuracy, precision, recall, and F1 answer different questions about errors.

The threshold depends on the decision

Missing a real warning and raising a false alarm can have different costs. Choose evaluation measures that reflect the purpose and audience.

Review 7: Models Can Learn More Complex Patterns

- **Week 15 — MLPs:** layers of simple units combine to represent more flexible patterns; backpropagation uses the chain rule to update weights.
- **Week 16 — PCA:** principal components summarize directions of variation, but a component is not automatically an explanation.
- **Week 17 — recurrent models:** a hidden state carries information from earlier time steps into a later prediction.

Complexity has a cost

More computation and memory do not guarantee better evidence, a fair comparison, or a more helpful decision.

Review 8: Research Is More Than a Model

- **Weeks 18–19:** read critically, check assumptions, and connect methods as parts of a system.
- **Week 20:** turn an idea into a bounded question, an evidence plan, a comparison, and a clear proposal.
- A model is a simplified representation. Its limits should match the limits of the claim.

Researcher's responsibility

Separate observed data, estimates, assumptions, and results. Never present a hoped-for benefit as a measured outcome.

Review 9: A Community Project Connects Evidence to Action

Data $\rightarrow$ Question $\rightarrow$ Analysis $\rightarrow$ Baseline $\rightarrow$ Decision $\rightarrow$ Check

- **Week 21:** audit a public dataset and identify the person who might use the result.
- Make a bounded question that the data can actually answer.
- Compare with a simple or current-practice baseline.
- Name a possible benefit and how someone could verify whether it occurred.

Review 10: The Questions That Travel with You

1. What is the question, and who cares about the answer?
2. What do the data represent, and what is missing?
3. What is the simplest fair comparison?
4. What evidence would change my mind?
5. What can I responsibly conclude or recommend?

The point of the review

These questions connect programming, mathematics, AI, literature research, and community value into one research habit.

Presentation 1: Begin with Your Audience

Before building slides, ask:

- Who is listening, and what do they already know?
- What do they need to understand, decide, or remember?
- Which terms need a plain-language explanation?

One project, two audiences

A researcher may want the model's error rate. A school facilities manager may first want to know which decision the analysis could help with and what evidence it uses.

Presentation 2: Open with a Human Question

An inspiring introduction gives the audience a reason to listen; it does not promise a result.

1. Begin with a vivid but accurate question, moment, or small example.
2. Name the real-world problem and who experiences it.
3. State your research question and what you set out to learn.

Example opening

"On a hot afternoon, when might a school use the most electricity—and what information would help staff plan?"

Presentation 3: Explain Purpose and Value Before Details

Give listeners a map before showing them technical steps:

1. **Purpose:** What question are you answering?
2. **Value:** Why might the answer matter, and to whom?
3. **Plan:** What evidence and comparison will you use?
4. **Details:** What method, result, and limitation should they see?

Value is a reason to investigate, not proof

Say what benefit is possible, who could receive it, and what would need to be checked. Do not claim the benefit has already happened.

Presentation 4: Give the Talk a Simple Story

Problem → Question → Evidence → Finding or plan → Meaning and limits

- Use one main message per slide and a title that says the point.
- Explain a graph's axes, units, and takeaway before moving on.
- Use signposts: "First. . . ," "The key comparison is. . . ," and "This means. . . ."
- Leave formulas and background details out unless they help answer the question.

Presentation 5: Speak So People Can Follow

Practice

- Face the audience; use notes as prompts, not a script to read.
- Slow down at important ideas and pause between sections.
- Project your voice and pronounce technical terms clearly.

Prepare

- Time one full rehearsal aloud.
- Check slide text and visuals from the back of the room.
- Rehearse the transition into and out of your talk.

Accessible speaking

Clear structure, readable visuals, and a steady pace help many listeners; no single speaking style is required.

Presentation 6: Be Ready for Questions

1. Listen to the full question and pause before answering.
2. Answer the question directly, then give the evidence or reason.
3. If you do not know, say so and name what you would check next.
4. Be ready to explain your baseline, a limitation, and one alternative.

A strong answer can include uncertainty

"Our data cannot answer that yet. We would need more evidence; for now, our conclusion is limited to the question we studied."

Opportunity Cost 1: Every Choice Leaves Alternatives

Definition

Opportunity cost is the value of the best alternative you give up when you choose an option.

- It is not every possible thing you did not choose; it is the most valuable feasible alternative.
- It can be time, learning, access, usefulness, or money—not only a financial cost.
- Giving something up is not automatically a mistake. Choices require priorities when resources are limited.

Opportunity Cost 2: A Research-Time Example

Suppose you have two afternoons and can investigate only one project first.

Option	Possible value	Constraint or trade-off
Study school energy	A school staff member may be able to use the result	Need permission and comparable meter data
Study EV charging	Explore a timely transportation question	Local charging records may not be public

If you choose school energy

The opportunity cost is the value of the best feasible alternative you set aside—perhaps the EV project, if that was your next-best choice. It is not automatically a bad outcome.

Opportunity Cost 3: Choose with Priorities and Constraints

1. List a few realistic project options, not every imaginable idea.
2. Set criteria that matter: access to data, time, safety, fit with a user's need, and ability to check results.
3. Identify hard constraints first; then compare the options that remain.
4. Select the strongest feasible project and name the best alternative you are giving up.

A priority is a reasoned choice

"This project is my first choice because it is useful to a specific user, the data are accessible, and I can test it within the time available."

Opportunity Cost 4: Explain What You Chose—and Why

A thoughtful research plan briefly answers:

- Which topic did I select, and what evidence makes it feasible?
- Which promising alternative did I not select?
- What constraint or priority drove that decision?
- What value or learning opportunity am I giving up, and what could make me reconsider?

Be fair to the alternative

Do not call the unchosen idea worthless. Explain why it is a weaker fit *for this project, now*, given the evidence and constraints.

Homework: Build a Literature-Based Baseline

Continue the community project you proposed in Week 21; do not start a new topic.

1. Find **three relevant research papers, technical reports, or authoritative project sources** and verify each source directly.
2. Fill the literature table: question, data/context, method, comparison or baseline, reported finding, limitation, and relevance to your plan.
3. Use the sources to describe what is already known and one specific gap or question that remains.
4. Define a simple baseline and a measurable outcome for your own plan; explain why your project is feasible and what alternative you set aside.

Evidence first

Cite real sources, distinguish their reported results from your expectations, and never invent a citation or finding. See `week22_homework.tex` for the worksheet.